Report contents

S.NO	Contents	Page.no
1	Introduction	1
2	Components	1
3	Theory	2
4	Circuit diagram	12
5	Implemented circuit	13
6	Features	14
7	Problems and solutions	16

Abstract:

This project presents the design and hardware implementation of a Digital Clock with Alarm functionality built entirely using fundamental digital integrated circuits (ICs), specifically the 74HC390 Dual Decade Counter, the CD4543 BCD-to-7 Segment Latch/Decoder/Driver, and the 74HC157 Quad 2-to-1 Data Selector/Multiplexer. This approach demonstrates the core principles of sequential and combinational logic design without relying on a microcontroller or complex programming.

Aim:

To design a clock integrated with alarm which is functional by using the basic decade counters.

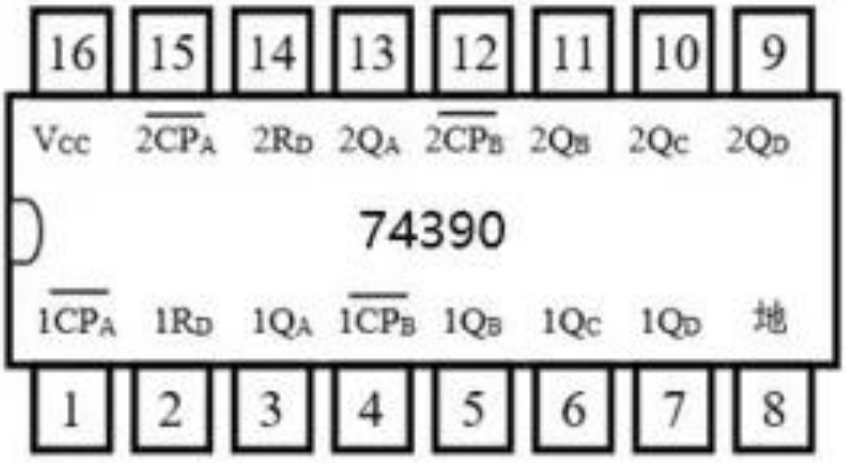
Components used:

Components	No of units
Ic 74HC390	6
Ic cd4543	6
Ic 74HC157	4
7-segment displays	6
Push buttons	24
1k ohm resistors	56
10k ohm resistors	20
1 micro farad capacitor	20
Ne555 timer	1
Pcb 18cm*30cm	1
Ic 74HC08	2
Ic 74HC32	2
Passive Buzzer	1
IC 74HC85	4
22 AWG single strand wires	Required

Theory:

IC 74HC390(Decade counter):

The 74HC390 is a versatile high-speed CMOS integrated circuit consisting of two independent 4-bit ripple counters. Each of the two counters is internally divided into a divide-by-two section and a divide-by-five section, each with its own clock input and and a shared asynchronous, active-HIGH Master Reset This dual structure allows for flexible counting modes: for a digital clock, the most common configuration is the BCD Decade (Mod-10) Counter achieved by connecting the div 2 output to the clock input and applying the main clock signal to the clock input This results in count sequence (0000 to 1001 BCD). By utilizing external logic to detect a specific count (like the BCD 6 for the tens digits of minutes/seconds) and driving the MR pin HIGH, the IC can be truncated to function as a Mod-6 counter, making the 74HC390 ideal for constructing the cascaded chain of a digital timepiece.

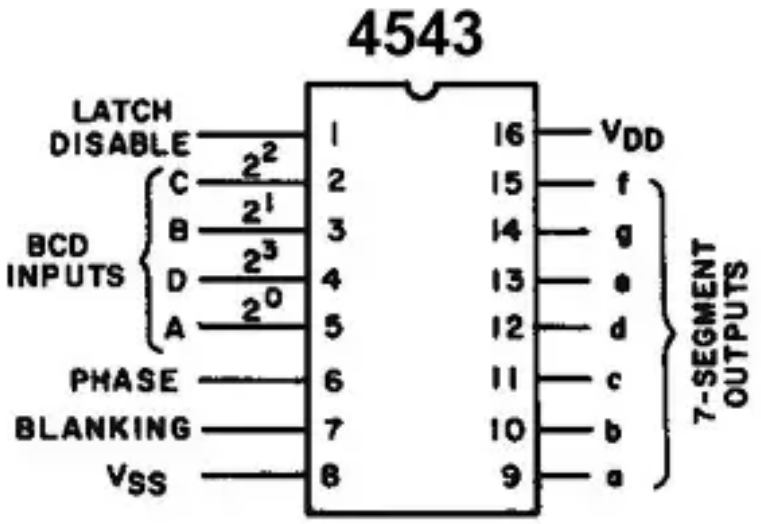


74390 芯片引脚图

In each IC we consist of two-decade counter so upon we use each counter for clock and another one for the alarm.

IC CD4543:

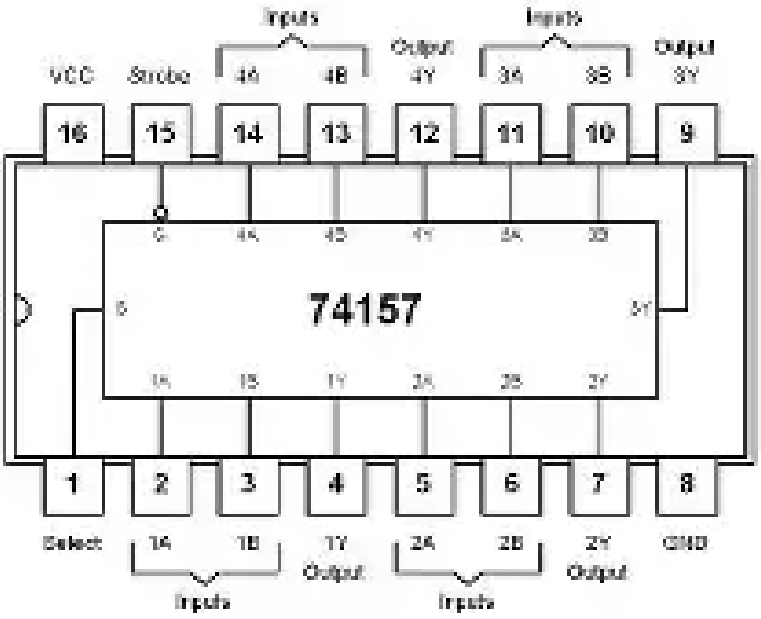
The CD4543 BCD-to-7 Segment Latch/Decoder/Driver is a CMOS integrated circuit that serves as the crucial interface between the binary counting logic (like the 74HC390) and the visual 7-segment display. Its primary function is to decode the 4-bit BCD input (representing digits 0-9) into the 7-segment output pattern needed to illuminate the correct display segments. A key feature is the Phase (PH) control pin, which allows the IC to drive both Common Cathode (PH=HIGH) and Common Anode (PH=LOW) LED displays, adding flexibility to the design. Furthermore, it incorporates a latch controlled by the Latch Disable (LD) pin that can freeze the current displayed number when LD is LOW, allowing the counting logic to continue operating without constantly updating the display. Finally, the Blanking BI pin provides a simple way to switch off the entire display (BI=HIGH) for power management or display multiplexing.



In this circuit we are using 6 drivers to decode bcd values to 7 segment values, it can drive common cathode, common anode 7 segment led.

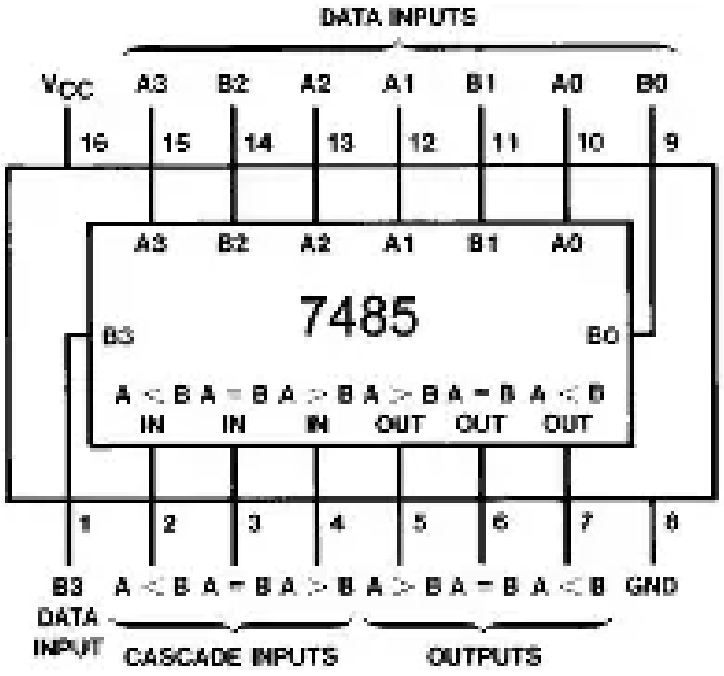
IC 74HC157:

The IC 74157 is a quad 2-to-1 line data selector/multiplexer belonging to the 74-series TTL logic family. It consists of four independent multiplexers, each capable of selecting one of two data inputs and directing it to a single output line. All four multiplexers share a common select input (S) and a common enable input (G). When the enable input (G) is low, the device is active; depending on the state of the select input (S), it outputs either the data from input C (when S = 0) or from input D (when S = 1). When the enable input is high, all outputs are forced to a low state, effectively disabling the device. The IC 74157 operates on a +5V supply and is compatible with TTL logic levels. It comes in a 16-pin dual inline package (DIP), where each of the four multiplexers has two data inputs, one output, and shares the common control lines. This IC is widely used in digital systems for data routing, signal selection, and parallel data transfer, such as choosing between two 4-bit binary inputs in microprocessor or memory interface applications. Its typical propagation delay is around 10 nanoseconds, making it suitable for high-speed digital circuits.



IC 74HC85:

The IC 7485 is a 4-bit magnitude comparator belonging to the 74-series TTL logic family. It is designed to compare two 4-bit binary numbers (A and B) and determine their relative magnitudes. The IC provides three output signals indicating whether A is greater than B (A>B), A is equal to B (A=B), or A is less than B (A<B). It can also be cascaded with other 7485 comparators to compare numbers larger than 4 bits by using the cascading inputs for higher-order comparisons. The device operates from a typical +5V DC supply and is TTL-compatible, making it suitable for use in microprocessor systems, arithmetic units, and digital sorting circuits. The IC has a 16-pin dual inline package (DIP) configuration, with inputs for the two 4-bit binary numbers (A0–A3 and B0–B3), three cascading inputs (A>B in, A=B in, A<B in), and three corresponding outputs. When used, if the inputs A and B are identical, the A=B output goes high; if A is greater, the A>B output goes high; and if A is smaller, the A<B output goes high. The 7485 is widely used in digital electronics for comparison-based operations such as sorting, arithmetic decision-making, and control logic, where determining numerical relationships between binary values is essential.



NE555 Timer:

The IC 555 timer (commonly known as the NE555) is one of the most popular and versatile integrated circuits used in electronics. It is designed primarily for timing, pulse generation, and oscillator applications. Developed by Signetics in 1972, the NE555 operates in three basic modes — monostable, astable, and bistable — making it extremely useful for various digital and analog circuits.

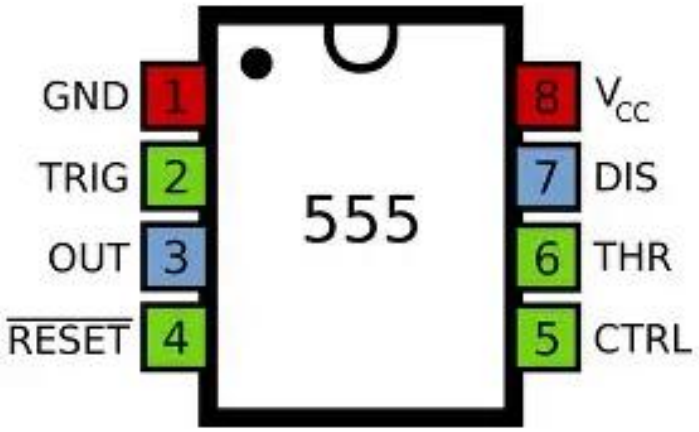
In the monostable mode, the 555 functions as a one-shot pulse generator, producing a single output pulse of a specific duration when triggered. In the astable mode, it acts as an oscillator that continuously generates a square wave output, used for clock pulses, blinking LEDs, and tone generation. In the bistable mode, the 555 acts as a flip-flop, where the output remains in one of two stable states until triggered to switch.

The IC operates typically with a DC supply voltage between 4.5V and 15V, making it compatible with both TTL and CMOS logic systems. It has 8 pins in a Dual Inline Package (DIP): Pin 1 (Ground), Pin 2 (Trigger), Pin 3 (Output), Pin 4 (Reset), Pin 5 (Control Voltage), Pin 6 (Threshold), Pin 7 (Discharge), and Pin 8 (Vcc). The timing interval or oscillation frequency is controlled by external resistors and capacitors, which set the charge and discharge times of an internal capacitor.

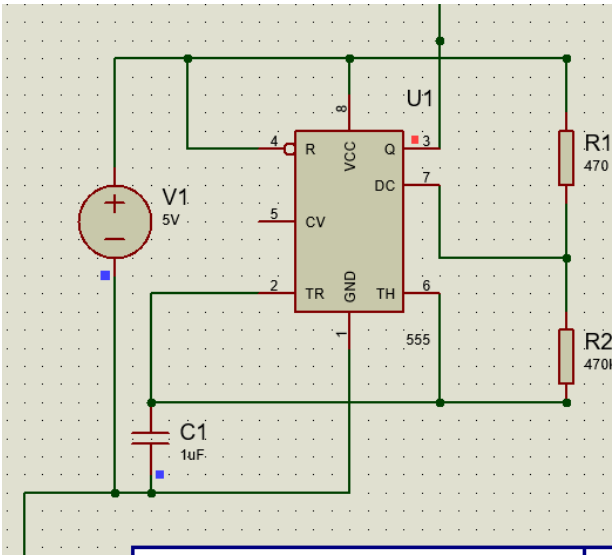
Internally, the 555 consists of two voltage comparators, a flip-flop, a discharge transistor, and a voltage divider made of three 5 kΩ resistors (which is why it's called the "555" timer). This internal structure allows it to generate stable time delays ranging from microseconds to hours, depending on the external components used.

The NE555 timer is widely used in applications such as pulse-width modulation (PWM), LED flashers, timed delays, tone generation, frequency generation, switch debouncing, and waveform generation. Its

simplicity, low cost, and reliability make it one of the most important ICs in both educational and industrial electronics.



Circuit used to produce 1Hz frequency with NE555 timer



Practically it produces on 1.2Hz which drives the seconds counter at 0.8s
It can be varied by environmental elements like temperature which varies the resistance and capacitance

7-segment display:

A 7-segment display is an electronic display device used to represent decimal numbers (0–9) and some alphabetic characters in digital systems. It is one of the most common output devices used in digital clocks, calculators, meters, and electronic counters to show numerical data in a simple, human-readable form.

The display consists of seven individual LED segments (labeled ‘a’ to ‘g’) arranged in a rectangular fashion to form the number eight (8). By selectively lighting specific segments, different digits and a few letters can be displayed. Many 7-segment displays also include an eighth segment (a dot), known as the decimal point (DP), used to represent fractional values.

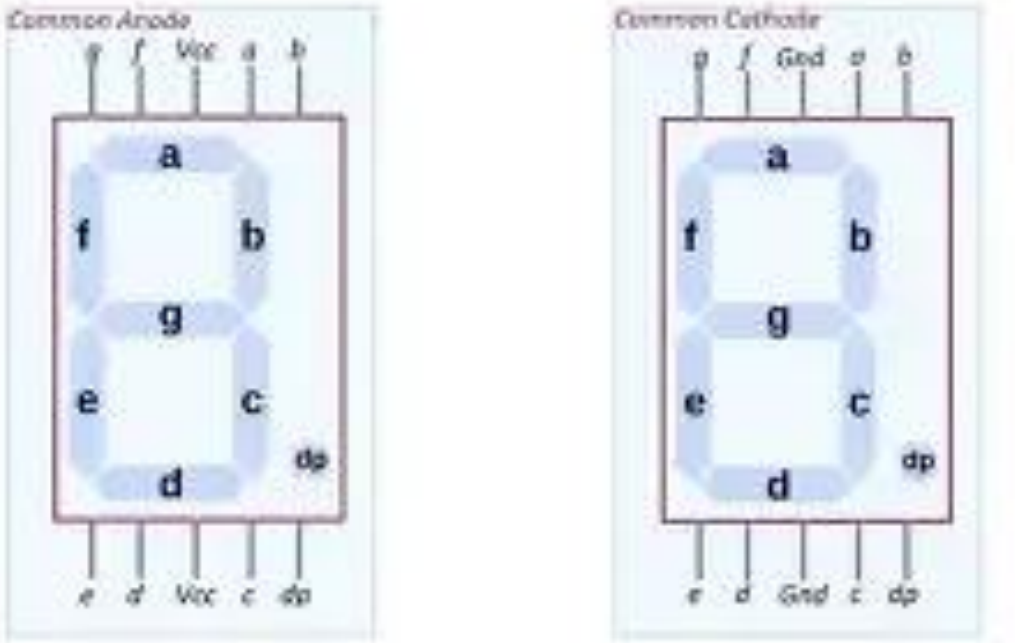
There are two main types of 7-segment displays based on their internal wiring:

1. Common Anode (CA): All the anodes (positive terminals) of the LEDs are connected together and supplied with a HIGH (+Vcc) voltage. To turn on a segment, the corresponding cathode is connected to LOW (0V).
2. Common Cathode (CC): All the cathodes (negative terminals) are connected together to ground (0V), and individual segments are turned on by applying a HIGH signal to their anodes.

Each segment is an LED, and when current flows through it, it emits light. The display can be driven directly from logic circuits, microcontrollers (like Arduino), or through decoder/driver ICs such as IC 7447 (for common anode) and IC 7448 (for common cathode), which convert binary-coded decimal (BCD) inputs into the required 7-segment outputs automatically.

In operation, for example, to display the digit “2,” segments a, b, g, e, and d are turned ON, while the others remain OFF. Similarly, by controlling different combinations of segments, digits from 0 to 9 can be displayed.

The 7-segment display is simple, inexpensive, and widely used in both educational and industrial electronic devices for numeric visual indication, though for more complex text or graphics, dot matrix and LCD/OLED displays are preferred.



Seven segment display pinout

Push button:

A push button using a pull-down resistor is a simple and essential circuit configuration used to provide a stable digital input signal (HIGH or LOW) to a microcontroller or logic circuit when a button is pressed or released.

In this setup, the push button is connected between the positive supply voltage (Vcc) and the input pin of the microcontroller, while a pull-down resistor is connected between the input pin and ground (GND). The pull-down resistor ensures that the input pin has a defined logic state (LOW) when the button is not pressed, preventing it from "floating" or picking up noise, which could cause erratic behavior.

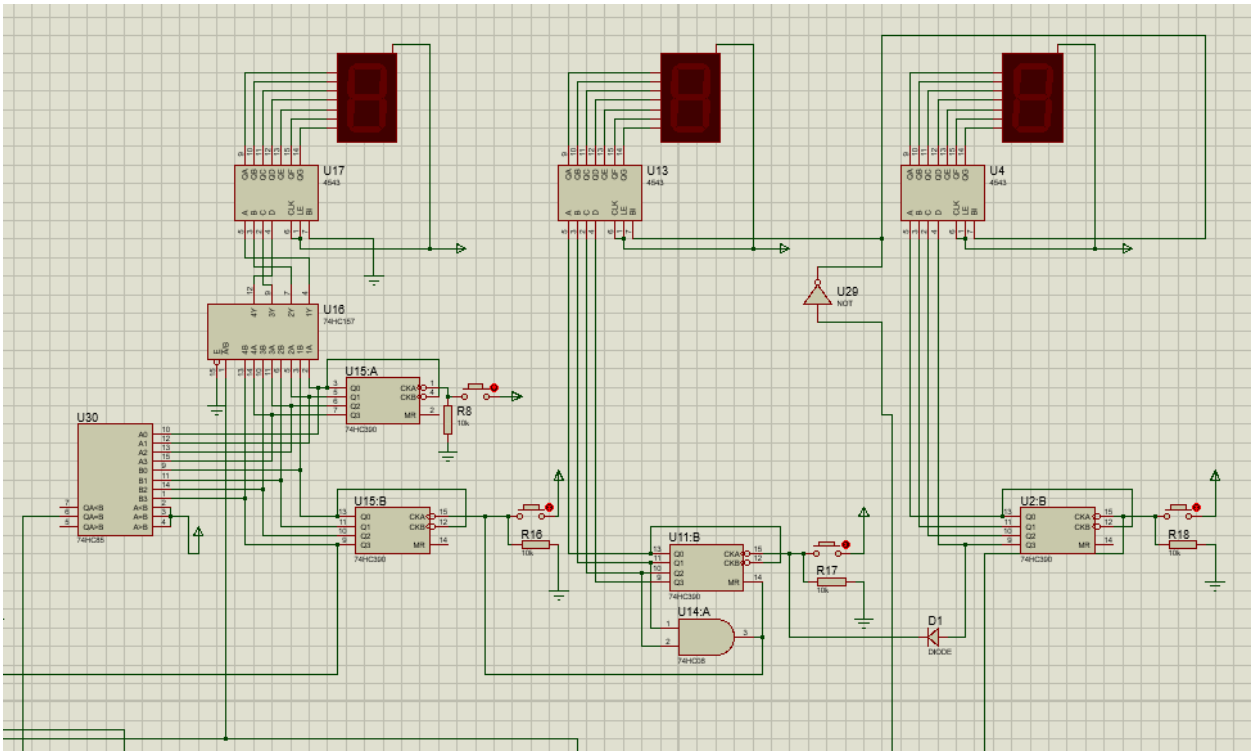
Here's how it works:

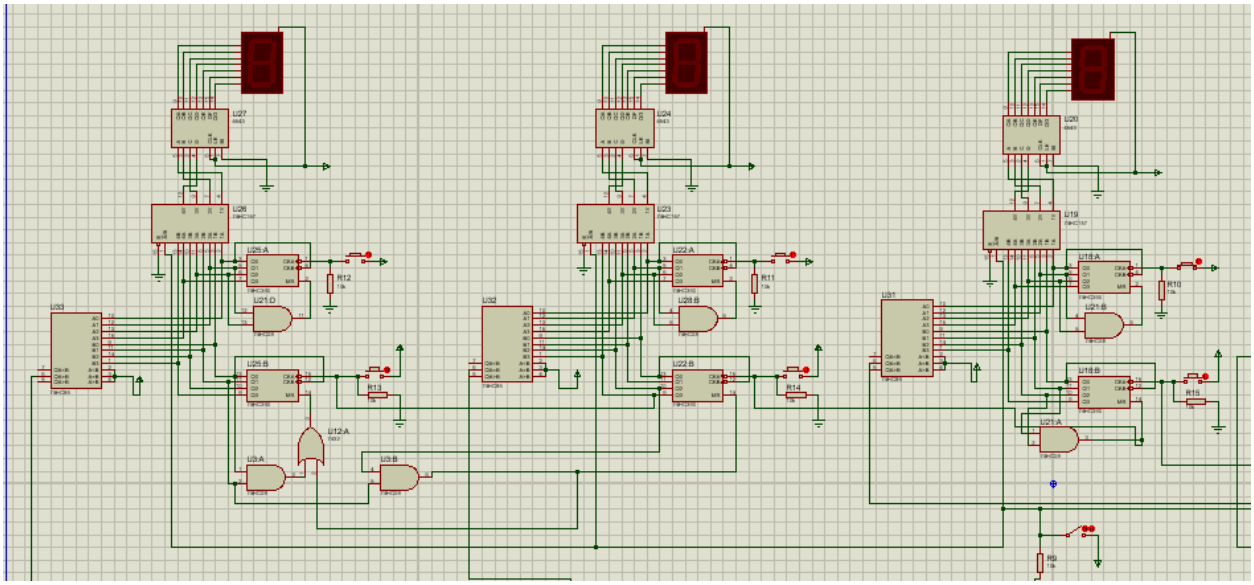
- When the button is not pressed, the circuit is open, and no current flows from Vcc to the input pin. The pull-down resistor (typically 10kΩ) connects the input pin to ground, keeping it at a LOW (0V) logic level.
- When the button is pressed, the circuit is closed, and current flows from Vcc to the input pin, overriding the resistor and bringing the input to a HIGH (logic 1) state.

This setup ensures clean, reliable transitions between logic states — LOW when released and HIGH when pressed — which can then be read by digital systems like Arduino, Raspberry Pi, or any microcontroller.

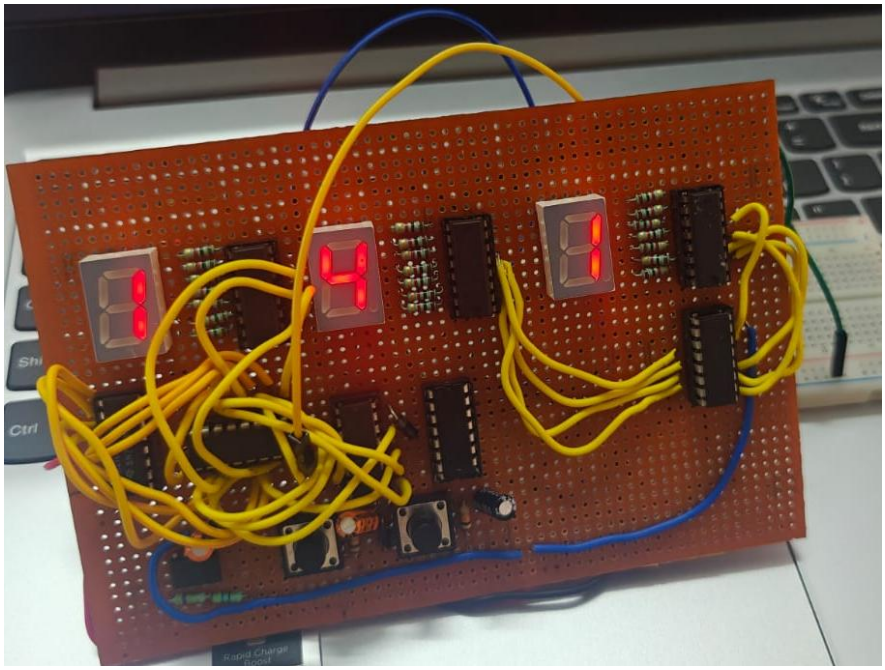
In summary, a pull-down resistor prevents floating inputs and ensures that the microcontroller receives a definite LOW signal when the button is not pressed. It is a basic yet crucial concept in digital input interfacing and embedded electronics.

Circuit diagram:





Circuit photograph:



Here are the features of your Digital Clock and Alarm Circuit using NE555 Timers, Counters, Comparator, and Multiplexer — presented clearly for a report or presentation:

1. **Accurate Time Generation**:
    - The NE555 timer IC is used in astable mode to generate precise clock pulses for the seconds counter.
    - Stable operation ensures accurate timekeeping.
2. **Cascaded Counter Design:**
    - Multiple decade counters (for seconds, minutes, and hours) are cascaded to form a complete digital clock.
    - Each counter stage drives a 7-segment display for clear time indication.

3. **Alarm Setting Feature:**
    - Separate alarm counters are provided to set the desired alarm time.
    - Push buttons are used to manually adjust the alarm hours and minutes.
4. **Comparator-Based Alarm Trigger:**
    - A comparator circuit continuously checks whether the clock time equals the alarm time.
    - When both match, it activates the buzzer, indicating the alarm condition.
5. **Shared 7-Segment Display with Multiplexing:**
    - The same 7-segment displays are used for both clock and alarm time.
    - A 2-to-1 multiplexer (MUX) allows switching between normal time display and alarm setting mode without stopping the clock.
6. **Manual Mode Switching:**
    - By changing the MUX selection line input, the user can easily toggle between clock display and alarm display.
7. **Buzzer Indication:**
    - An audio buzzer provides an audible signal when the alarm time is reached.
8. **User-Friendly Operation:**
    - Simple push-button interface for time and alarm setting.

- Compact circuit design suitable for practical and educational applications.

9. **Expandability:**
    - The design can be extended to include additional features like AM/PM indication, digital calibration, or temperature display.

Further implementation:

This circuit offers ample scope for future enhancements to make it more user-friendly and feature-rich. Additional functionalities such as AM/PM indication can be incorporated to clearly distinguish between morning and evening hours, enhancing the clock's usability in 12-hour format systems. A snooze feature can also be implemented, allowing the user to temporarily silence the alarm for a few minutes before it automatically reactivates, improving convenience. Furthermore, a timer or countdown mode can be added, enabling the circuit to function not only as a clock and alarm but also as a multifunctional timekeeping device. These upgrades can be easily integrated with minor hardware modifications and additional control logic, making the design more versatile and practical for real-world applications.

Problems Faced in Implementation

During the implementation of the circuit, a few challenges were encountered that affected the overall progress and testing. One of the primary issues was the unavailability of the required ICs, which caused delays in assembling and testing the complete system. Additionally, voltage fluctuations in the power supply led to unstable circuit behavior, causing irregular counting and display errors in the timer and counters. These fluctuations affected the reliability of the output and required frequent circuit adjustments. Ensuring a stable power source and timely

availability of components would help achieve more consistent and accurate performance in future implementations.

Solutions to problems:

To overcome the issues faced during implementation, several practical measures can be adopted. The problem of IC unavailability can be resolved by sourcing components in advance through reliable electronic suppliers or by using equivalent IC substitutes that perform the same function. Maintaining an inventory of essential components can also prevent project delays. To address voltage fluctuations, a regulated power supply should be used to ensure stable operation of all ICs and display modules. Adding decoupling capacitors across the power lines of ICs can further filter out minor voltage variations and noise. Additionally, incorporating a backup battery or voltage stabilizer can help maintain consistent performance during power interruptions. By implementing these solutions, the circuit can achieve greater stability, reliability, and accuracy in its operation.

Result

The designed digital clock and alarm circuit functioned successfully as intended. The clock operated accurately, displaying the time correctly through the 7-segment displays, while the alarm feature also worked effectively. When the set alarm time matched the current clock time, the comparator circuit activated the buzzer, providing a clear and reliable alarm indication. Overall, the circuit demonstrated proper synchronization between the clock and alarm units, confirming the successful implementation of the design.